

Syllabus/Course Outline
Multivariable Calculus (Calculus II)
Fall Semester 2024

Multivariable Calculus (CS3)**Semester: Fall 2024****Credit Hrs: 3****Instructor: Ali Haider****Pre-requisite: Calculus & Analytical geometry (MTH 106)****Classroom:****Code: MTH 111****Contact Hrs: Sec A, Lecture Tue & Thru (2:00 – 3:30 pm, 3:30 – 5:00 pm)****Sec B, Lecture Mon & Wed (3:30 – 5:00 pm, 12:30 – 2:00 pm)****Sec C, Lecture Tue & Thru (11:00 – 12:30 am, 12:30 – 2:00 pm)****E- mail: haider.ali1@bnu.edu.pk****Course Introduction**

This course is continuation of calculus-1 course. The main objective of the course is to make students proficient in techniques and skills of multivariate and vector calculus. The emphasis will be on application of learned techniques to the solution of problems of science and engineering. The students will learn necessary mathematical skills with an understanding of basic concepts and a working knowledge of applications. It is hoped that, on completion of this course, the students will be trained enough to apply these mathematical tools to solve problems in their area of specialization.

Course Contents

- **Parametric Equations and Polar Coordinates:** Parametric representation of plane curves; Calculus with parametric curves; Polar coordinates; Graphing polar equations; Conic sections in polar coordinates; Areas and arc lengths in polar coordinates.
- **Vector Algebra:** Vectors in three dimensions; Dot and cross product; Lines and planes; Three dimensional quadric surfaces.
- **Functions of Several Variables:** Limits and continuity; Partial derivatives; Increments and differentials; Chain rules; Directional derivatives, gradient; Extrema of functions of several variables; Relative extrema.
- **Multiple Integrals:** Double integrals, definition, and evaluation; Area and volume; Double integrals in polar coordinates; Surface area; Triple integrals in Cartesian.
- **Integrals and Vector Field:** Line integrals, Green's theorem, Stoke's theorem, Divergence theorem.

Mapping of CLOs to PLOs

CLO No	Course Learning Outcomes	PLOs	Blooms Taxonomy
CLO 1	Apply calculus of parametric and polar curves to solve engineering problems related to areas, arc lengths and conics, and understanding of vectors and their manipulation.	PLO 1	C3 (Applying)
CLO 2	Demonstrate the understanding of analytic geometry of three dimensions and geometric surfaces of linear, quadratic functions, directional derivatives, and extreme values of functions	PLO 1	C3 (Applying)
CLO 3	Demonstrate the understanding and apply multivariate integral calculus of scalar and vector functions to various applied problems	PLO 1	C3 (Applying)
CLO 4	Demonstrate understanding of Green's theorem, Stoke's theorem, divergence theorem.	PLO 1	C3 (Applying)

CLO Assessment Mechanism

Assessment tools	CLO 1	CLO 2	CLO 3	CLO 4
Quizzes	25%	25%	25%	25%
Assignments	5%	5%	5%	5%
Midterm Exam	60%	50%		
Final Exam	10%	20%	70%	70%

Overall Grading Policy

Assessment Items	Percentage
Announced Quizzes	20%
Assignment Quizzes	10%
Midterm Exam	30%
Final Exam	40%

Text and Reference Books

Textbook:

- “Thomas' Calculus” by George B. Thomas, Jr., Maurice D. Weir, Joel R. Hass. 13th Edition 2015. Pearson, USA.

Reference Books:

- “Calculus: Early Transcendentals” by James Stewart. 6th Edition 2008. Brooks/Cole USA.
- “Calculus” by Swokowski, Olinick, Pence. 6th Edition 1994. PWS, USA.
- “Calculus and Analytic Geometry” by Thomas et al 9th Edition, USA.

Administrative Instructions

- According to institute policy, 75% attendance is **mandatory** to appear in the final examination.
- Assignments must be submitted as per instructions given by the course instructor or mentioned in the assignments.
- Students may work on home assignments in collaboration with each other, but they must submit their own work; no copying from others. Violation of this will adversely affect their quiz/exam results.

Computer Usage

Students are encouraged to solve some assigned homework problems using the available software such as Python and MATLAB.

Lecture Breakdown

Lecture	Topic	Chapter
01	Parametric representation of plane curves. (11.1)	Chapter 11
02	Calculus with parametric curves. (11.2)	Chapter 11
03	Polar coordinate system. (11.3)	Chapter 11
04	Some important polar curves & Sketching of simple polar curves. (11.4)	Chapter 11
05	Conic sections in rectangular coordinates. (11.6)	Chapter 11
06	Conic sections in polar coordinates. (11.7)	Chapter 11
07	Area and arc length in polar coordinates. (11.5) Q_1, A_1	Chapter 11
08	Vectors in two & Three dimensions and problems (12.1, 12.2)	Chapter 12
09	Scalar product (12.3)	Chapter 12
10	Vector product (12.4)	Chapter 12
11	Analytic Geometry of planes & Straight lines in three dimensions. (12.5)	Chapter 12
12	Cylinders & Important Quadric surfaces. (12.6)	Chapter 12
13	Functions of several variables. (14.1)	Chapter 14
14	Limit of functions of several variables. (14.2-a) Q_2, A_2	Chapter 14
15	Continuity of functions of several variables. (14.2-b) Mid- Term	Chapter 14

16	Partial derivatives and related questions. (14.3)		Chapter 14
17	Increments, differentials, and chain rules. (14.4)		Chapter 14
18	Directional derivatives and gradient. (14.5)		Chapter 14
19	Tangent planes and normal planes. (14.6)		Chapter 14
20	Extrema of function of several variables. (14.7)		Chapter 14
21	Double integrals. (15.1)	Q_3, A_3	Chapter 15
22	Area and volume by double integrals. (15.2)		Chapter 15
23	Area and volume by double integrals. (15.3)		Chapter 15
24	Triple Integrals in Rectangular Coordinates. (15.5)		Chapter 15
25	Line Integrals. (16.1)		Chapter 16
26	Vector Fields and Line Integrals: Work, Circulation, and Flux. (16.2)		Chapter 16
27	Path Independence, Conservative Fields, and Potential Functions. (16.3)		Chapter 16
28	Green's Theorem in the Plane. (16.4)		Chapter 16
29	Surfaces and Area. (16.5)		Chapter 16
30	Surface Integrals (16.6)	Q_4, A_4	Chapter 16
31	Stokes' Theorem. (16.7)		Chapter 16
32	The Divergence Theorem and a Unified Theory. (16.8)		Chapter 16

Note: This course outline is designed on the premise of pre-requisite course calculus-1, according to HEC policy, supervised by Dr. Shafaat Ahmed and Dr. Shazia. This outline and lecture distribution serves only as rough guidance of the course. It may be changed or modified as and when deemed necessary by the instructor. The instructor is at liberty to best distribute the number of lectures and/or change the sequence of topics to cover the entire course.